

Summary: Performing Quantum Computation with Classical Optics

Xiaofeng Qian, Son H. Nguyen, Matthew Daddino, Ching Man Wong

Physics Department, Stevens Institute of Technology, Hoboken, NJ 07030, USA

Abstract

Quantum computers promise transformative capabilities for molecular simulation and optimization problems, yet practical implementations face significant technological barriers. This research demonstrates an alternative approach: implementing quantum algorithms using classical optical systems. We present a framework for performing quantum computation through optical components, specifically realizing the Variational Quantum Eigensolver (VQE) algorithm for molecular energy calculations. Our implementation encodes qubits using photonic polarization states and spatial modes, leveraging wave plates (quarter-wave and half-wave), beamsplitters, and dove prisms to construct quantum gates. We detail the theoretical decomposition of the UCCSD (Unitary Coupled Cluster Single and Double excitations) ansatz for the hydrogen molecule, deriving the necessary rotations and entangling operations. The optical realization employs Jones calculus to systematically map quantum gates onto polarization-dependent optical transformations, with Stokes vector measurements enabling quantum state readout. This hybrid classical-optical approach provides insights into quantum circuit implementation while leveraging the well-established optical laboratory infrastructure, demonstrating that quantum algorithms can be executed through photonic platforms as an alternative to superconducting or trapped-ion systems.

Introduction

Following O'Malley et al., we begin with the reduced two-qubit electronic Hamiltonian for molecular hydrogen:

$$\hat{H} = g_0\mathbb{I} + g_1Z_0 + g_2Z_1 + g_3Z_0Z_1 + g_4Y_0Y_1 + g_5X_0X_1, \quad (1)$$

where X_i , Y_i , and Z_i are Pauli operators acting on qubit i , and the coefficients g_k depend on molecular geometry (in particular, bond length). This compact form captures the essential electronic structure terms needed for ground-state estimation in a near-term quantum setting.

Estimating the minimum eigenvalue of Eq. 1 is the central task of the Variational Quantum Eigensolver (VQE): prepare a parameterized trial state, measure the expectation value $\langle \hat{H} \rangle$, and iteratively update parameters to approach the ground-state energy. In the standard digital model, this procedure is implemented with a reference state preparation, single-qubit rotations, and entangling gates that realize a UCC-inspired ansatz for H_2 .

This thesis investigates how the same quantum-information workflow can be implemented using classical optics. We map qubit states to photonic degrees of freedom (polarization and spatial mode), and map unitary gate operations to physically realizable optical transformations built from wave plates, beam splitters, and mode-control elements. Jones-matrix and Stokes-parameter formalisms are used to connect the theoretical circuit operations to measurable optical quantities.

The goal is not to claim a universal quantum speedup with classical hardware, but to construct a physically transparent and experimentally accessible platform for studying quantum-circuit structure, variational optimization behavior, and measurement strategies motivated by quantum chemistry. By grounding the study in the O'Malley Hamiltonian and the H_2 VQE benchmark, this work provides a concrete bridge between quantum-algorithm design and optical laboratory realization.

Preparation of the Quantum Circuit

To implement the VQE algorithm for the Hamiltonian in Eq. 1, we first need to prepare the quantum circuit that corresponds to the UCCSD ansatz for H_2 . This involves decomposing the unitary operator into a sequence of single-qubit rotations and two-qubit entangling gates.

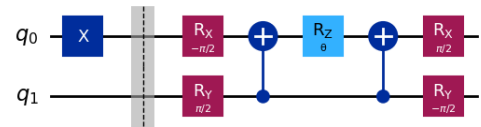


Figure 1: UCCSD ansatz circuit for the hydrogen molecule benchmark used in this work.

Optical implementation of Quantum Gates